

Health & Safety

Signaling standards and vehicle-pedestrian protection systems



The vehicle marshalling and vehicle-pedestrian protection system standards, described in this document, are a practical guide to reinforce the effective and safe management of construction site vehicles and to mitigate the potential for uncontrolled pedestrian interface.

This document does not supersede the instructions within the the reference document **Standard Vehicles/Pedestrian Interaction (BYTP-REF-H&S-2212)**.

The information within this document can be used in support of pre-start briefings where there is the potential for vehicle-to-vehicle collisions and interface with pedestrians, that must be controlled whilst undertaking workplace activities.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

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1. Life-Saving rules

RISK COLLISION

- I respect the **traffic segregation lanes** for vehicles and pedestrians
- I do not go into the work zone of moving plant or vehicles without the **operator's acknowledgement**
- I do not enter an **exclusion zone**

I DO NOT GO INTO THE WORK ZONE OF MOVING PLANT OR VEHICLES WITHOUT THE OPERATOR'S ACKNOWLEDGEMENT

I RESPECT THE TRAFFIC SEGREGATION LANES FOR VEHICLES AND PEDESTRIANS

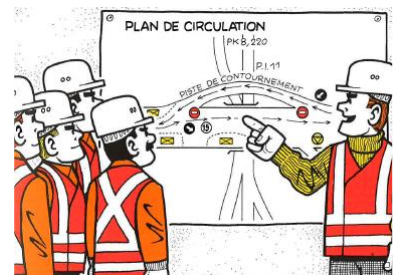
If unable to adhere to these rules, I "STOP" and alert my hierarchy.

SAFETY CULTURE

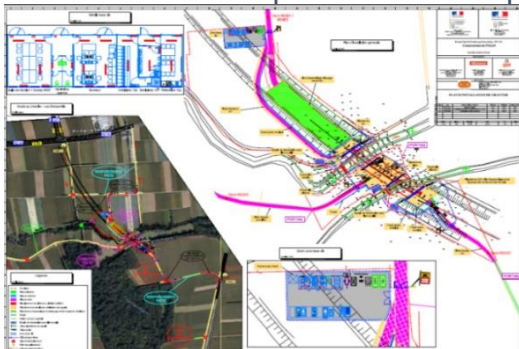
2. Organisation of traffic on Construction site

The **Traffic Management Plan (TMP)** is in addition to the '*Site Installation Plan*', detailing site set up and organisation of the site, considering the scale of the site, specifically:

- Direction of traffic, one-way flow of traffic where possible;
- Areas dedicated to turn vehicles (turning zones);
- Trained vehicle marshal to support reversing manoeuvres
- Risk assessment must support the use of vehicle marshals
- Dedicated & safe plant & vehicle parking areas;
- Dedicated pedestrians walkways with physical barriers.



Note: the 'TMP' is presented to employees during the site induction.



Site installation plan used as the background

Magnets of machines mobilized on the site at T time

3. Vehicle/Plant Marshal

In situations where it is not possible to use physical segregation between construction site vehicles and pedestrians, the planned use of a competent vehicle marshal is acceptable.



The vehicle marshal must:

- Be trained & familiar with the work environment
- Assume responsibility for the safe movements of site vehicles & plant, in the vicinity of pedestrians
- Dons the stipulated, task-specific PPE
- Be issued with adequate means of communication
- Have the necessary authority to enforce the established traffic plan.

The support of a vehicle marshal are mandated in the following situations:

- Where the risk assessment dictates the need for assisted management of vehicle movements, e.g., the potential for uncontrolled pedestrian interface;
- Reversing manoeuvres in environments where vehicles & plant are operational;
- Working environments with restricted visual fields (operators & pedestrians);
- Where physical segregation of vehicles/plant-pedestrian is not possible;
- In the absence of technical support, e.g., reversing cameras, means of detection.

4. Authorisation and Permits

- Compliant driving license is compulsory for all personnel who drive site vehicles, e.g., vans, flatbeds
- Compulsory driving authorisation must be issued by the employer (or organisation representative) for all vehicle/plant, based on:
 - An assessment of theoretical and practical skills
 - Of medical fitness
- A new driving authorization is issued annually and/or on the expiry date of the driving certificates in order to ensure their renewal.



RECTO

	
Driving authorization	
Ref :	Délivré par :
Nom :	Signature :
Qualité :	
Date :	

VERSO

Photo du titulaire	Le titulaire : Nom :
	Qualification :
	est autorisé à conduire les chariots
Catégorie	Limite de validité

5. Equipment conformity

- All construction plant and vehicles must comply with local regulations, including the manufacturer's safe use requirements
- All construction plant and vehicles are subjected to the regulatory thorough examination criteria, e.g., 12 months for lifting equipment and, 6 months for equipment used for lifting people, e.g. MEWPs
- Non-compliant, defective, or plant and equipment that lack a report of thorough examination are immediately quarantined.



6. Preparation & maintenance of pedestrian walkways

- Pedestrian pathways must be physically segregated from vehicles/construction plant movements, paying attention to dedicated pedestrian crossing points
- Prioritise information signage and lighting for pedestrian pathways near vehicle access roads and crossing points
- The materials used to construct pedestrian walkways within construction areas must comprise:
 - Materials that promotes a dedicated pedestrian walkways, e.g., coloured zones for visual reference
 - Rubber/resin gratings
 - A grain size of the materials ≤ 0.6 in order to avoid the risk of twisting ankles
- Installation of pedestrian gantries crossing worksite tracks, supported with information directional signs
- Pedestrian walkways and crossing points must be free from obstruction and storage of equipment is strictly prohibited.



Installation of red carpets on concrete paths



Flow separation



Pedestrian gantry crossing the construction site track



Directional signs



Electrical cables crossing protected paths

7. Physical Segregation & Information Signage

Should there be potential for the interface with the public, the traffic order (*document issued by traffic manager*) is briefed to employees and displayed at the work site to inform of the traffic rules on site and when egressing site onto public highways.

Information signage, e.g., erected signs and road markings, in support of the physical protection systems are carefully planned and adapted to the types of lanes, visibility, volumes of traffic and are positioned by competent person(s).



Reinforced concrete barrier



Modular lane barrier



Respect the distances provided between the protection systems and the form weights



Modular lane barriers are subject to deflection in the event of impact with a vehicle. This deflection is taken into account and validated by the Methods Dept., considering the location, the type of vehicle and the use of pedestrians

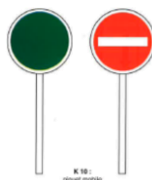
8. Temporary site information signage



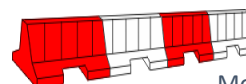
Access equipment for vehicles in worksite areas



Danger signs



Information signaling



Modular lane barrier (no containment level)



Cone



Pedestrian crossing

Position signage

9. Task-specific & Pedestrian Lighting

- When natural lighting is insufficient and unsafe, you must install lighting that is adapted to the work task to be carried out and or, the movement of pedestrians:
- Traffic routes/lanes (40 lux minimum)
- Stairway and storage areas (60 lux minimum)
- Work premises, changing rooms and toilets (120 lux minimum)
- Blind premises assigned to permanent work (emergency lighting)
- Additional lighting and specific PPE can be implemented according to local requirements.



Illuminating balloon



Portable projector



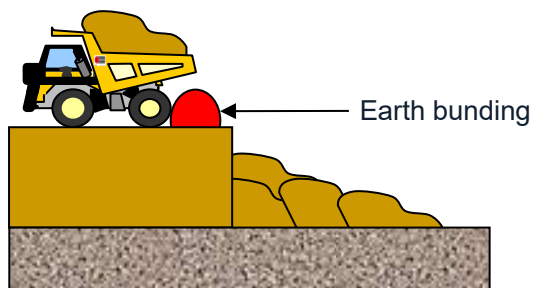
Specific PPE



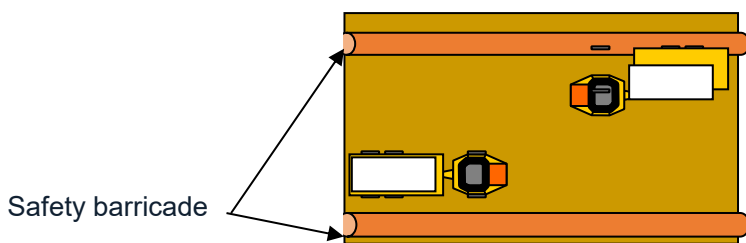
Led signaling strip

Prioritise LED lights

10. Stop (Earth) Bunding & Safety Barricades



Earth bunds designed to stop trucks at the edge of the embankment



Safety barricade at wheel level placed along the track to prevent machines leaving the track

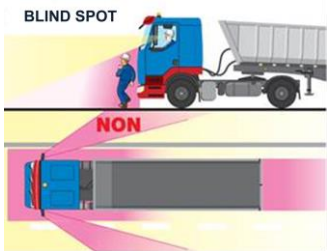
11. Vehicular ‘Blind spot’ detection systems

The Site Installation Plan or Traffic Plan must make it possible to avoid reversing maneuvers as much as possible.

In all cases, each machine is equipped with a beep and a reversing beacon.

The machines are also equipped with reversing camera(s) or radar(s):

- Depending on the type of machine used AND
- Depending on the risk analysis specific to the site concerned (coactivity, restricted spaces, etc.).

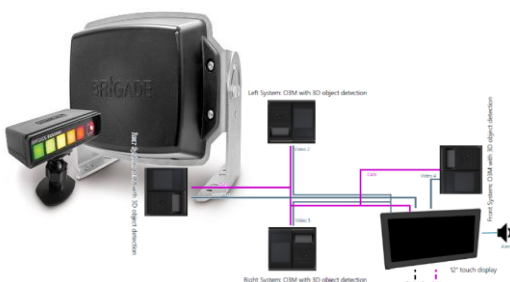


In the event of technical impossibility of equipping the machine, the systematic use of a vehicle marshal for any movement of the machine in reverse is essential.

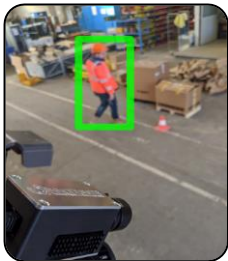
Examples of blind spot detection systems (non-exhaustive list):



Blaxtair



Backsense



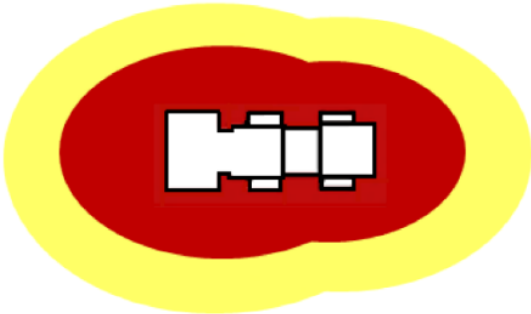
Innodura Artificial Intelligence

12. Detection system with servo control

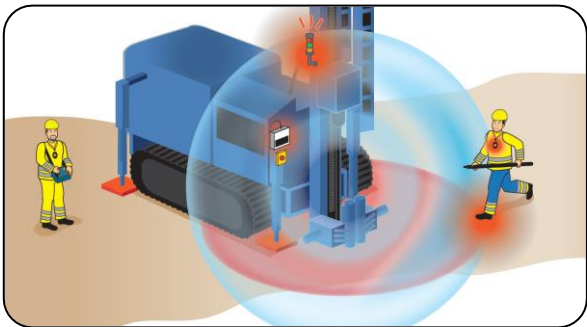
Servo detection is a system that alerts and interacts with machinery in a dangerous zone.

The system establishes a magnetic field around the machine. One or more generators can be mounted.

The detection system creates a ‘**WARNING ZONE**’ and a ‘**DANGER ZONE**’.



This field can encompass the entire machine and its turning radius, or be specific to certain working areas.



If the pedestrian, wearing a badge specific to the system, enters the pre-defined zone around the machine, an audible/vibrating and/or light alert is triggered.

In the event of a risk of collision, entrapment... the driver is warned by an audible and visual alarm.

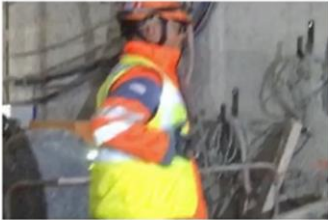
13. Pedestrian detection system by RFID tag/chip

Each pedestrian, wearing a specific badge, is detected by the vehicle's on-board detection system.

In the event of a risk of collision, the driver is warned by an audible and visual alarm, and the pedestrian wearing the badge by an audible and vibrating alert.



In-cab alert



Alert on the carrier



Alert on the machinery



14. Moving the multi-directional cherry pickers

The movement of the multidirectional cherry pickers is carried out with the pendulum raised in order to limit blind spots.

This also makes it possible to increase the speed of movement of the machine and its ability to cross ramps and slopes.



Warning – Movement in the lowered position is possible for reasons of space constraints and depending on the condition of the traffic lane and its unevenness.

Moving on this configuration coupled with the use of a vehicle marshal is not recommended.

Consider moving in reverse as it reduces the blind spots when travelling.